

Thyroid op 설명

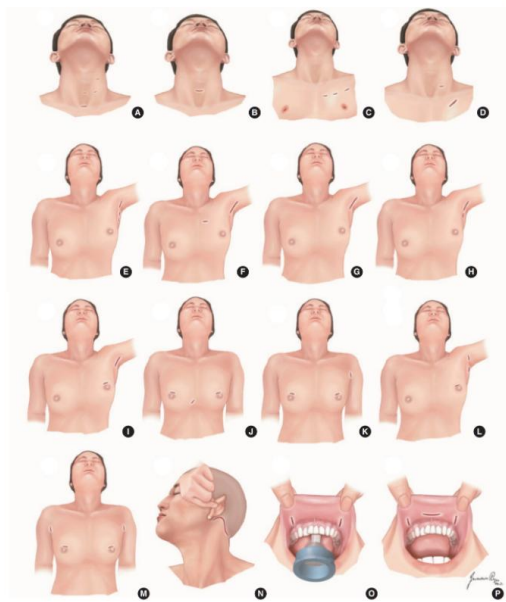
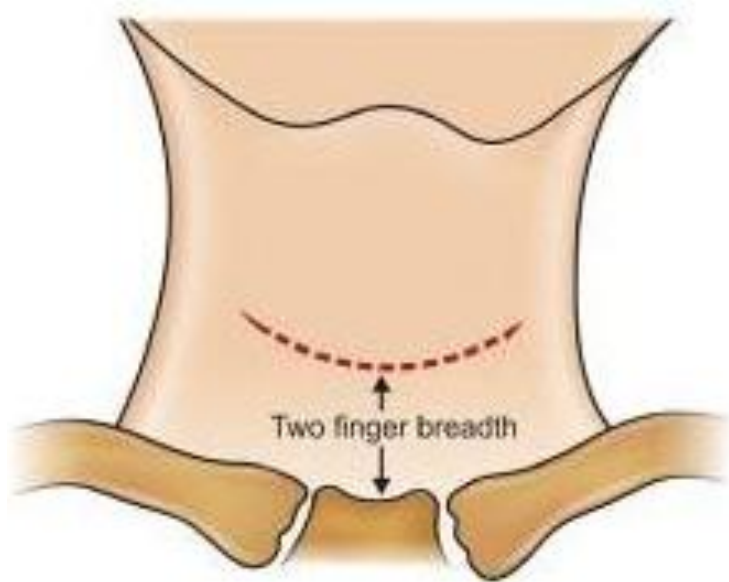


Fig. 1. Design of incisions in various robotic/endoscopic thyroidectomies. (A) Cervical approach with CO₂ insufflation. (B) Minimally invasive video-assisted thyroidectomy. (C) Anterior chest approach with CO₂ insufflation. (D) Video-assisted neck surgery. (E) Axillary approach with CO₂ insufflation. (F) Gasless transaxillary approach with anterior chest port. (G) Single-incision transaxillary approach. (H) Gasless unilateral axillary approach. (I) Gasless unilateral axillo-breast approach with CO₂ insufflation. (J) Breast approach with CO₂ insufflation. (K) Axillo-bilateral breast approach with CO₂ insufflation. (L) Unilateral axillo-breast approach with CO₂ insufflation. (M) Bilateral axillo-breast approach with CO₂ insufflation. (N) Postauricular facelift approach. (O) Transoral sublingual and vestibular approach with CO₂ insufflation. (P) Transoral vestibular approach with CO₂ insufflation.

<https://www.e-ceo.org/journal/view.php?doi=10.21053/ceo.2018.00766>



<https://epomedicine.com/wp-content/uploads/2014/06/thyroidectomy-incision.jpg>

OP fee?

ROBOT – 비보험

입원 기간 – 원칙적 1주

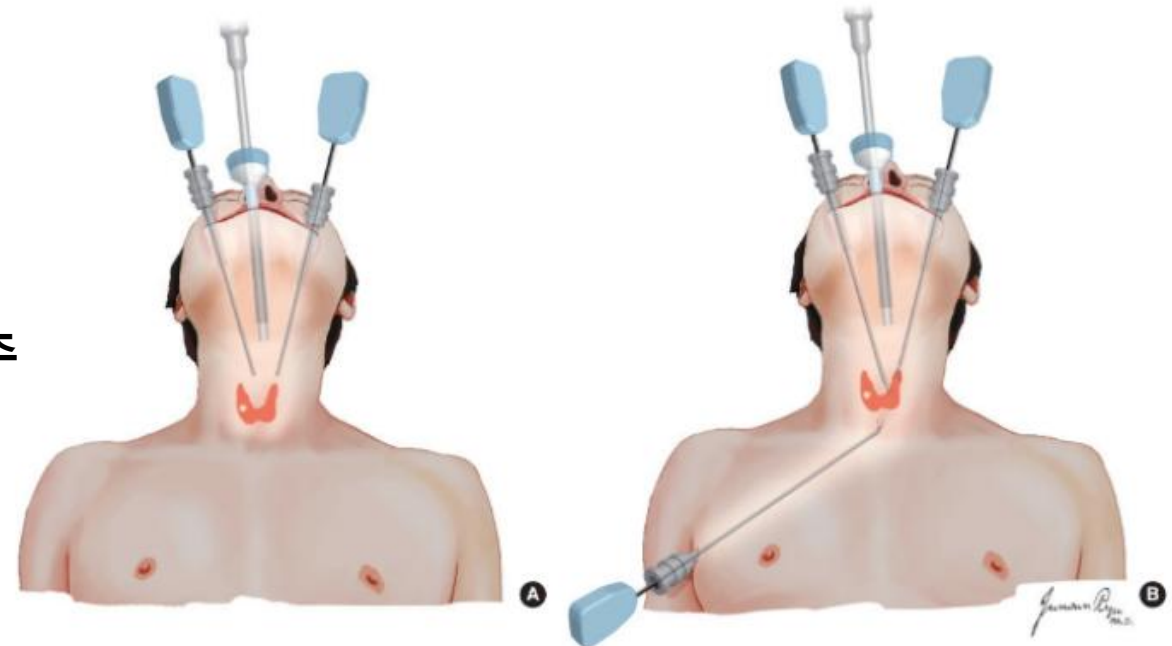


Fig. 5.

The transoral vestibular approach with CO₂ insufflation. (A) After creating a working space, a 30-degree robotic endoscope and two robotic instruments, such as bipolar Maryland forceps and monopolar scissors, are placed on either side of the endoscope. (B) Prograsp or Cardinal forceps is inserted through the right axillary port if necessary.



[https://www.journalofsurgicalresearch.com/article/S0022-4804\(20\)30475-3/fulltext](https://www.journalofsurgicalresearch.com/article/S0022-4804(20)30475-3/fulltext)



<https://www.sciencedirect.com/science/article/pii/S1521690X19300296>

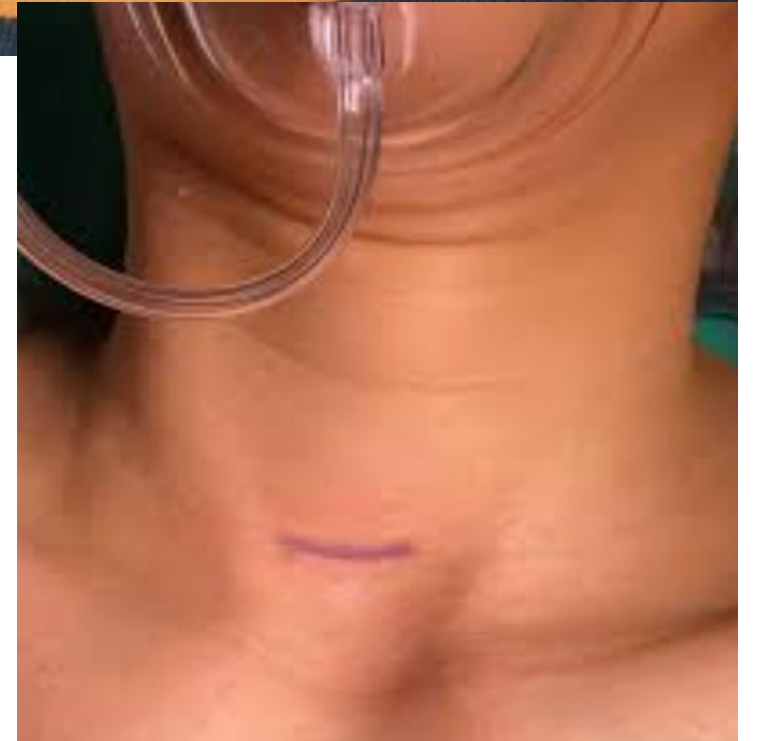
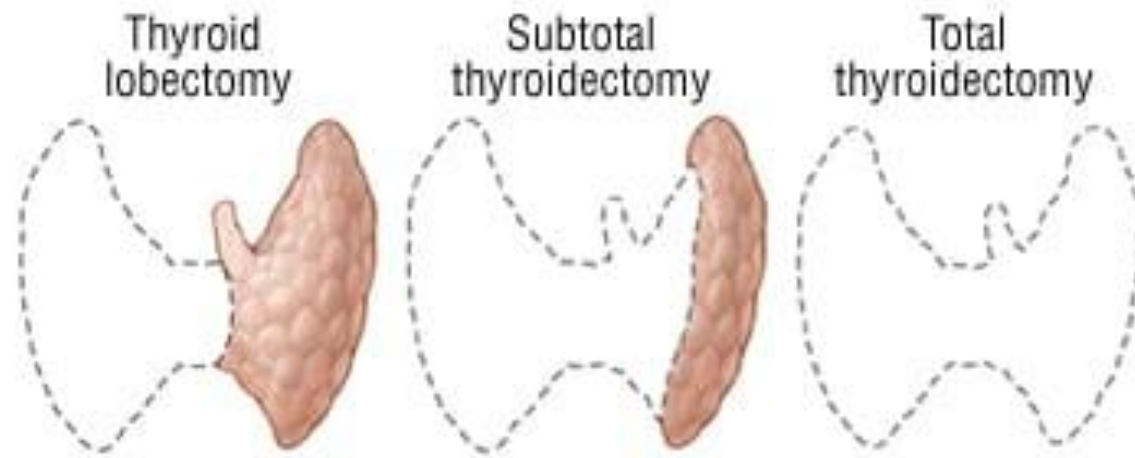
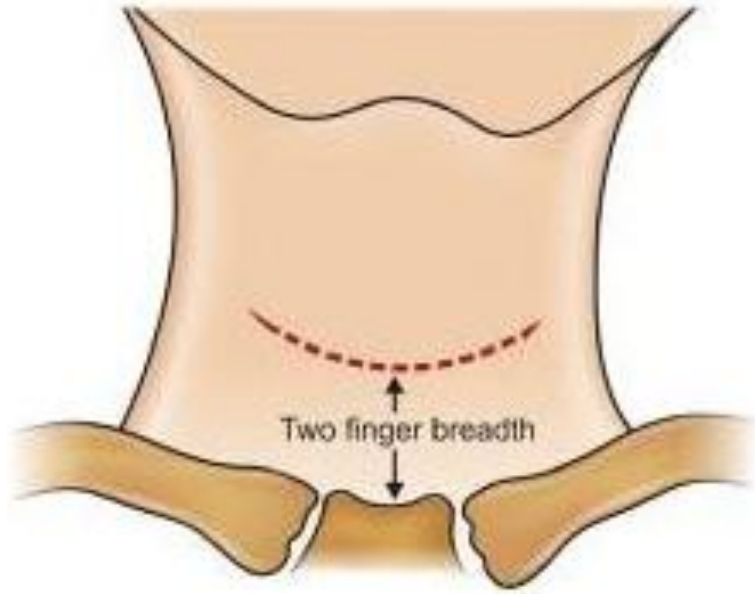
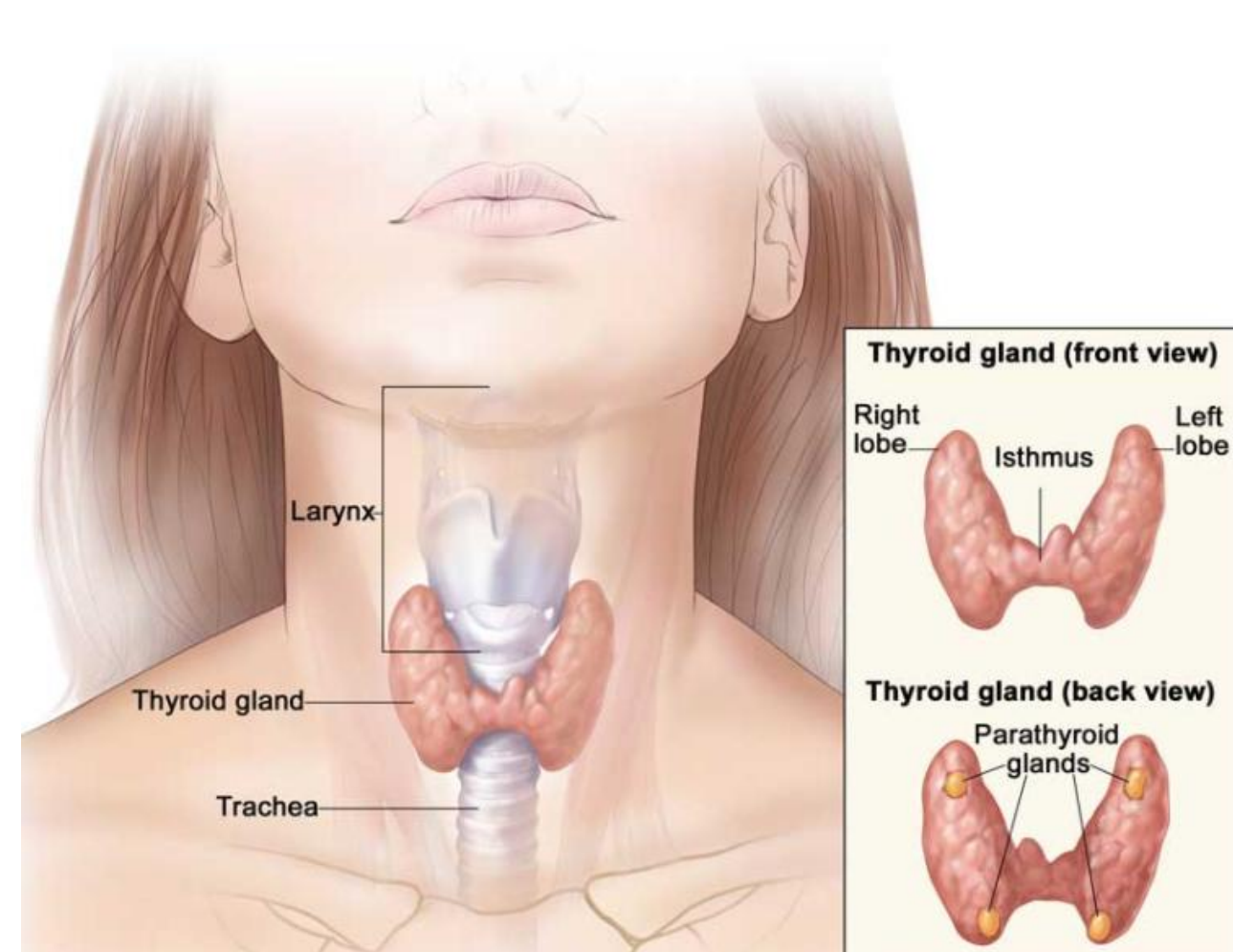
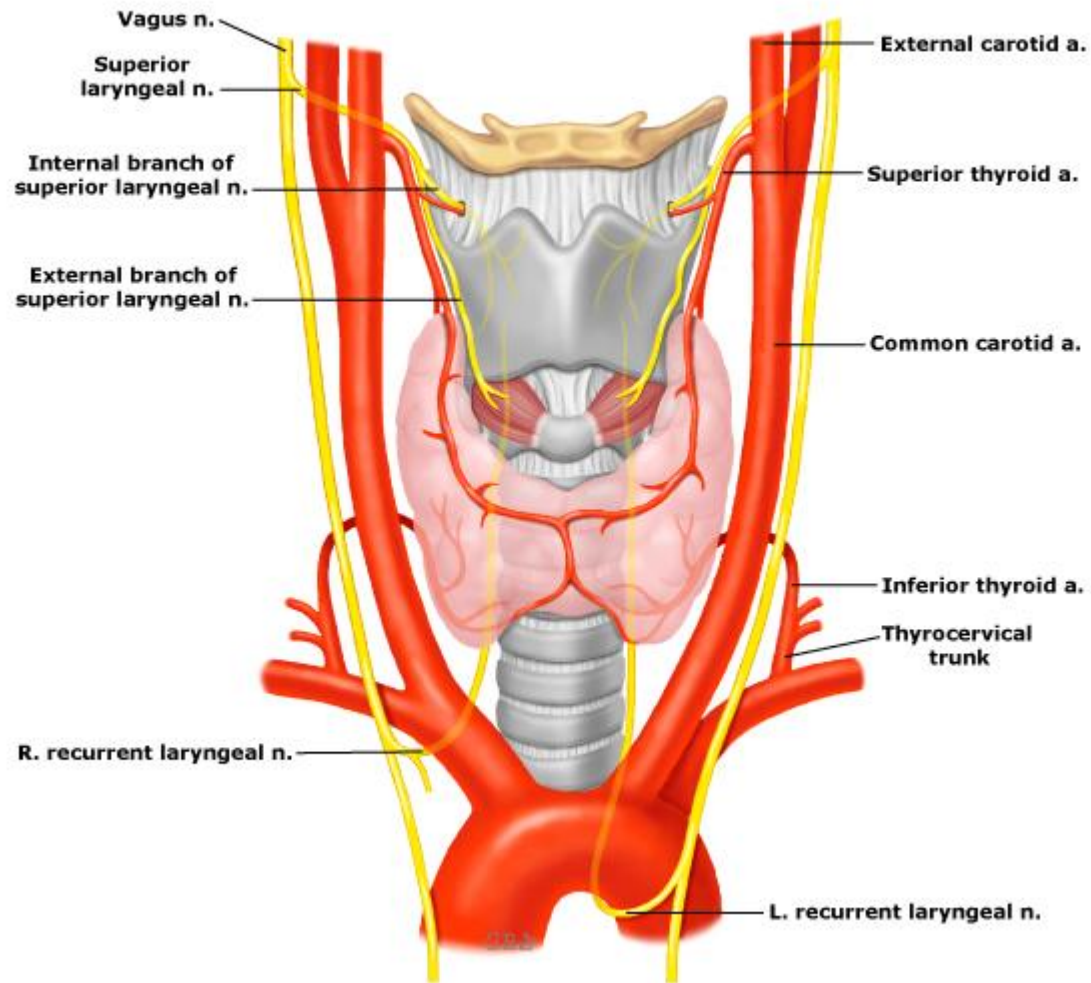


Table 1 Original framing of the Bethesda system showing different diagnostic criteria, risk of malignancy, and recommendation

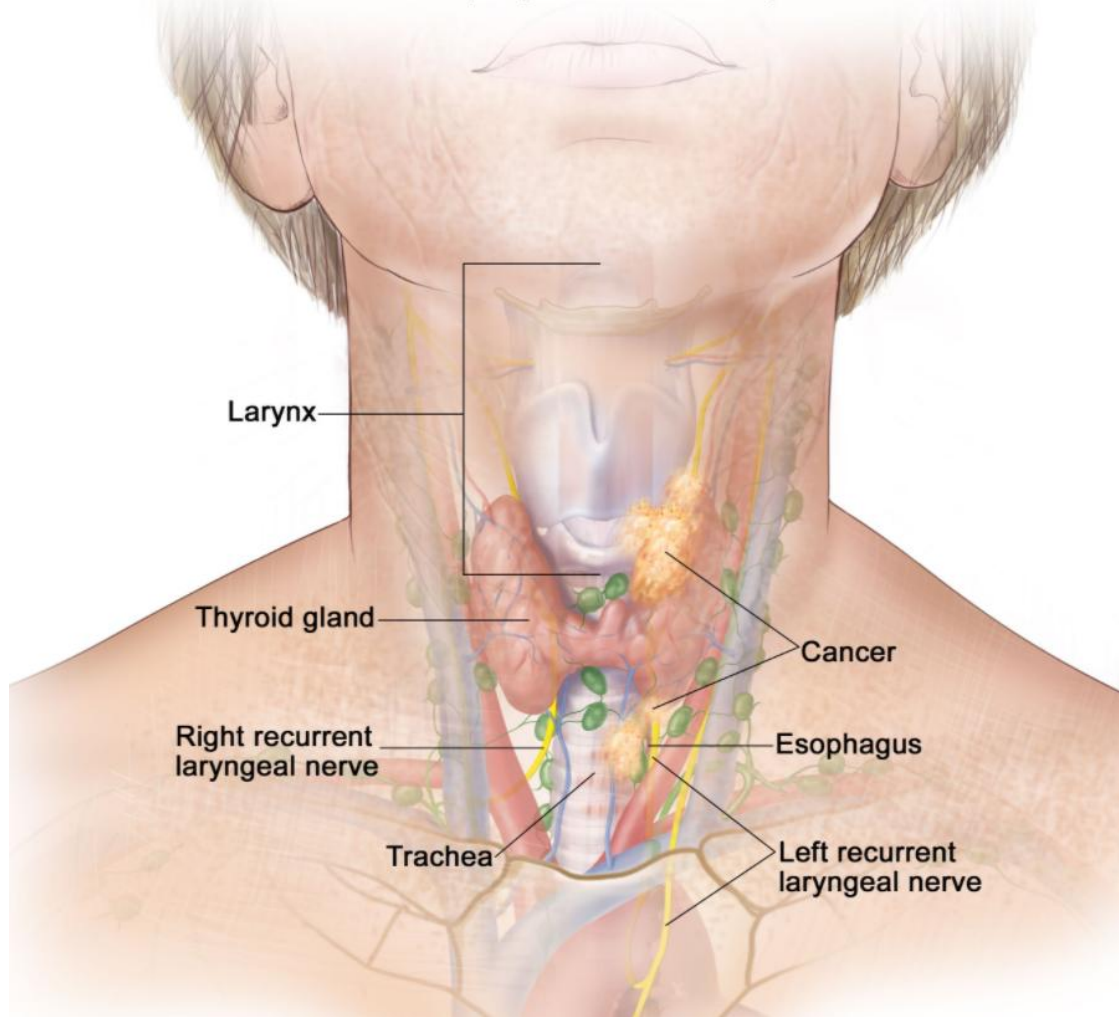
Bethesda category	Definition	ROM (%)	Recommendation
I. Nondiagnostic or unsatisfactory (ND)	Cyst fluid only;virtually acellular specimen; other (obscuring blood, clotting artifact, etc.)		Repeat USG guided FNAC
II. Benign (BN)	Consistent with a benign follicular nodule (includes adenomatoid nodule, colloid nodule, etc.); consistent with lymphocytic (Hashimoto) thyroiditis in the proper clinical context; consistent with granulomatous (subacute) thyroiditis;others	0–3%	Clinical follow-up
III. Atypia of undetermined significance or follicular lesion of undetermined significance (AUS)		5–15%	Repeat FNAC under image guidance
IV. Follicular neoplasm (FN) or Suspicious for a follicular neoplasm (SFN)		15–30%	Surgical lobectomy
V. Suspicious for malignancy (SM)	Suspicious for papillary carcinoma; suspicious for medullary carcinoma; suspicious for metastatic carcinoma; suspicious for lymphoma; others	60–75%	Near-total thyroidectomy or surgical lobectomy
VI. Malignant (MGT)	Papillary thyroid carcinoma;poorly differentiated carcinoma; medullary thyroid carcinoma; undifferentiated (anaplastic) carcinoma; squamous cell carcinoma;carcinoma with mixed features; metastatic carcinoma;non-Hodgkin lymphoma;others	97-99%	Near Total Thyroidectomy

Abbreviations: FNAC, fine needle aspiration cytology, USG, ultrasonography.

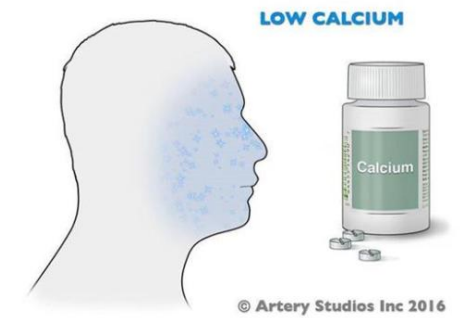
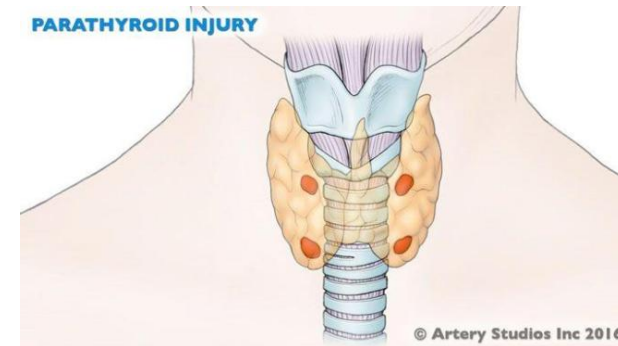
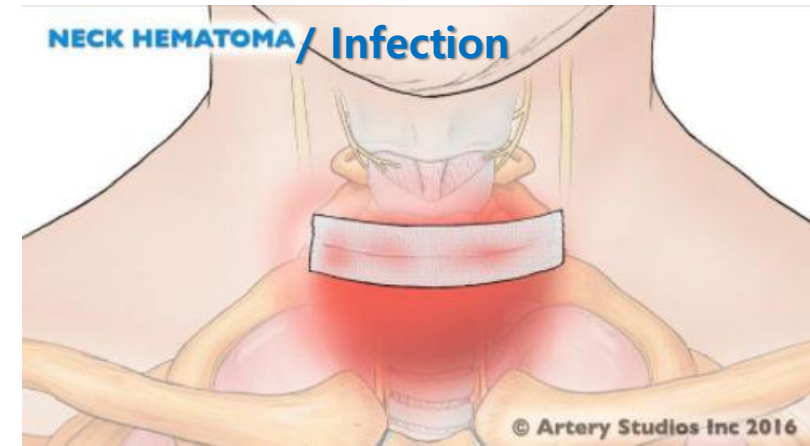
Nerve supply of the thyroid gland



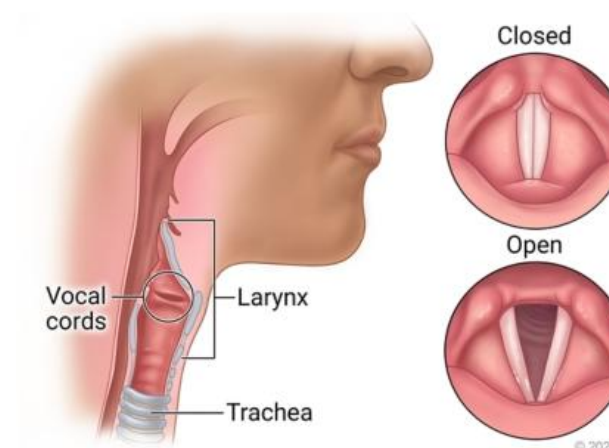
Stage III Papillary and Follicular Thyroid Cancer (55 years and older)



<https://nci-media.cancer.gov/pdq/media/images/779382.jpg>

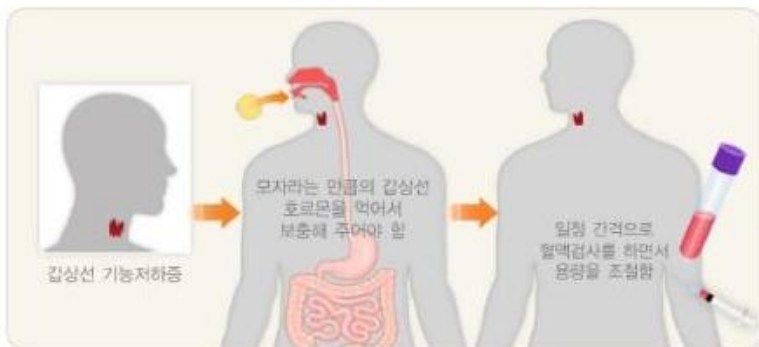
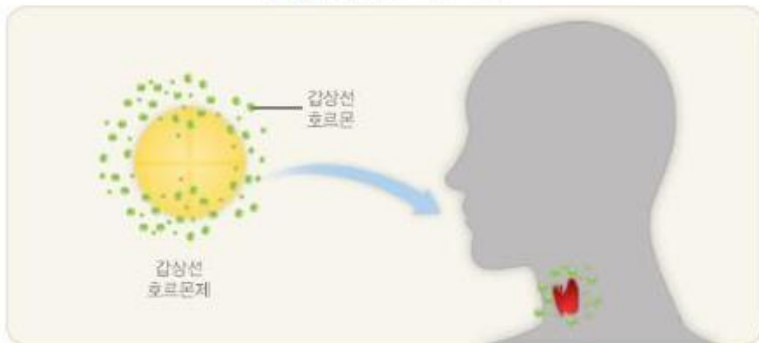


<https://www.myheadandnecksurgery-thyroid.ca/what-are-the-risks-of-my-surgery.php>



<https://www.cigna.com/individuals-families/health-wellness/hw/normal-vocal-cords-zm6118>

〈그림. 갑상선호르몬 치료〉

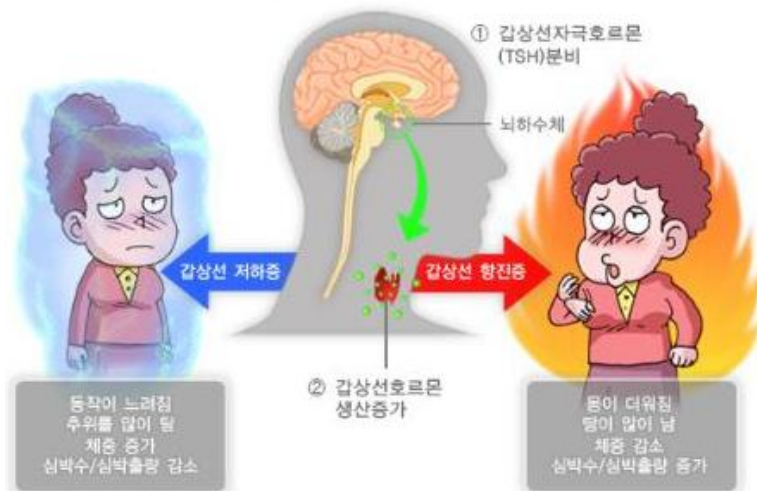


〈갑상선 저하증이 있는 경우〉



〈영구적 갑상선 저하증〉

〈그림. 갑상선기능저하증과 항진증〉



〈대사량 감소〉

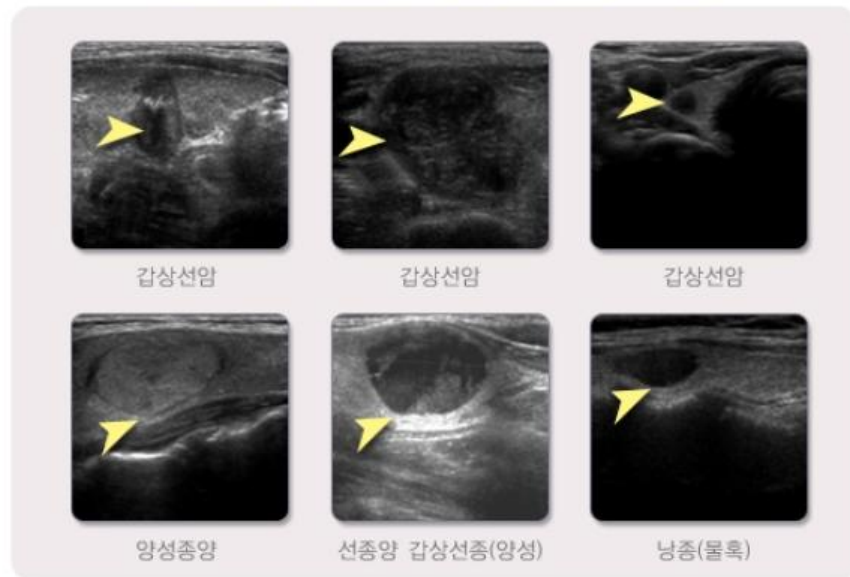
〈대사량 증가〉

이름	사진
Synthroid 0.025mg tab	
Synthroid tab 0.0375mg	
Synthroid 0.05mg tab	
Synthroid 0.075mg tab	
Synthroxine 0.088mg	
Synthroid 0.1mg tab	
Synthroid tab 0.15mg	

〈그림. 갑상선 초음파검사〉



〈각종 결절성 갑상선질환에서 갑상선 초음파 검사 소견들〉



출처: 국가건강정보포털

TNM definitions (AJCC 8e) for papillary, follicular, poorly differentiated, Hürthle cell, medullary, and anaplastic thyroid carcinomas	
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
T1	Tumor ≤ 2 cm in greatest dimension limited to the thyroid
T1a	Tumor ≤ 1 cm in greatest dimension limited to the thyroid
T1b	Tumor > 1 cm but ≤ 2 cm in greatest dimension limited to the thyroid
T2	Tumor > 2 cm but ≤ 4 cm in greatest dimension limited to the thyroid
T3*	Tumor > 4 cm limited to the thyroid or gross extrathyroidal extension invading only strap muscles
T3a*	Tumor > 4 cm limited to the thyroid
T3b*	Gross extrathyroidal extension invading only strap muscles (sternohyoid) from a tumor of any size
T4	Includes gross extrathyroidal extension into major neck structures
T4a	Gross extrathyroidal extension invading subcutaneous soft tissues, larynx, trachea, esophagus, or recurrent laryngeal nerve from a tumor of any size
T4b	Gross extrathyroidal extension invading prevertebral fascia or encasing carotid artery or mediastinal vessels from a tumor of any size
NX	Regional lymph nodes cannot be assessed
N0	No evidence of regional lymph nodes metastasis
N0a*	One or more cytologic or histologically confirmed benign lymph node
N0b*	No radiologic or clinical evidence of locoregional lymph node metastasis
N1*	Metastasis to regional nodes
N1a*	Metastasis to level VI or VII (pretracheal, paratracheal, or prelaryngeal/Delphian, or upper mediastinal) lymph nodes; this can be unilateral or bilateral disease
N1b*	Metastasis to unilateral, bilateral, or contralateral lateral neck lymph nodes (levels I, II, III, IV, or V) or retropharyngeal lymph nodes
M0	No distant metastasis
M1	Distant metastasis
*all categories may be subdivided as solitary tumor (s) and multifocal tumor (m) – the largest tumor determines the classification	

Staging guide for thyroid cancer (AJCC 8e)					
Age at diagnosis	T category	N category	M category	Stage	Expected 10-yr DSS
<i>Differentiated thyroid cancer</i>					
<55 years	any T	any N	M0	I	98–100%
	any T	any N	M1	II	85–95%
≥ 55 years	T1	N0/NX	M0	I	98–100%
	T1	N1	M0	II	85–95%
	T2	N0/NX	M0	I	98–100%
	T2	N1	M0	II	85–95%
	T3a/T3b	any N	M0	II	85–95%
	T4a	any N	M0	III	60–70%
	T4b	any N	M0	IVA	< 50%
	any T	any N	M1	IVB	< 50%
<i>Medullary thyroid cancer</i>					
any	T1	N0	M0	I	
	T2	N0	M0	II	
	T3	N0	M0	II	
	T1-3	N1a	M0	III	
	T4a	any N	M0	IVA	
	T1-3	N1b	M0	IVA	
	T4b	any N	M0	IVB	
	any T	any N	M1	IVC	
<i>Anaplastic thyroid cancer</i>					
any	T1-T3a	N0/NX	M0	IVA	
	T1-T3a	N1	M0	IVB	
	T3b	any N	M0	IVB	
	T4	any N	M0	IVB	
	any T	any N	M1	IVC	